

(b)

Random split (sounding-level)

random
split



$R^2 = 0.82$
(inflated)

test days sit between training days: same-day soundings share
weather and orbit state, so the random split leaks

fold-local calibration block
(early stop + conformal)

Date-blocked 5-fold CV (date_kfold)

fold 1



fold 2



fold 3



fold 4



fold 5



held-out R^2
0.53 ocean
0.39 land

2016 2017 2018 2019 2020 2021

116 training dates (semi-monthly, 2016–2020)



train



calibration



held-out test